

Systematic $xR+yC$ Structure Selection for RC-Based Building Digital Twins: Accuracy–Complexity Trade-Off and Practical Identifiability Considerations Under Real Operational Data

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Abstract. Digital Twins (DTs) in the built environment increasingly rely on reduced-order thermal models to enable continuous monitoring and operational forecasting under limited digital documentation. Resistance–Capacitance (RC) grey-box networks offer a compromise between physical structure and computational efficiency, yet selecting an appropriate $xR+yC$ topology remains largely empirical in practice. This paper proposes a structured comparison workflow for $xR+yC$ RC model selection as a lightweight thermal core for operational DT deployment. A bank of candidate RC structures is constructed and calibrated using one year of operational data from an existing institutional building (PLEIADData, Block A), based on hourly indoor temperature, outdoor temperature, and an effective HVAC thermal input. Candidate structures are compared using predictive accuracy metrics (RMSE/MAE) and penalized parsimony criteria (AIC/BIC-like scores) to explicitly explore the accuracy–complexity compromise under real-world excitation conditions. Results show that beyond a moderate structure order, additional RC states do not yield systematic validation gains, while increasing over-parameterization risk in lumped formulations. Full-year analysis reveals seasonal bias when a single global parameter set is applied across regimes, whereas short-horizon tracking remains acceptable in more stable winter conditions. Overall, the study provides a transparent, deployment-oriented approach to selecting a parsimonious RC topology under imperfect operational data, and clarifies practical limitations related to seasonal regime changes and lumped-parameter ambiguity in RC-based DT cores.

Keywords: Digital Twin, RC grey-box model, $xR+yC$ structure, model selection, identifiability, accuracy–complexity trade-off, building operation.

1 Introduction

Digital Twins (DTs) are increasingly positioned as enabling technologies for performance monitoring, optimisation, and decision support in the built environment. While early DT concepts originated in manufacturing, their extension to buildings has emphasized continuous synchronization between physical assets and their digital representations, enabling data-driven operational

optimization, predictive maintenance, and energy performance improvement [1, 2, 3]. In building operation and maintenance (O&M), energy efficiency remains one of the primary motivations for DT deployment [4, 5]. However, the implementation of full-scale DTs in existing buildings is often limited by interoperability issues, incomplete digital documentation, and the high modelling effort required for detailed physics-based simulations [2, 6].

In this context, reduced-order thermal modelling provides a pragmatic alternative. Resistance–Capacitance (RC) thermal networks constitute a class of grey-box models in which simplified physical structure is combined with parameter identification from measured data [7, 8, 9]. Unlike white-box simulation models, RC networks require limited geometric and material information, and unlike purely data-driven black-box models, they preserve physical interpretability of thermal states and parameters. This balance makes RC networks particularly attractive as computationally efficient predictive cores for operational Digital Twins.

RC models are commonly described using the notation $xR+yC$, where x and y denote the number of lumped thermal resistances and capacitances defining the structure [9]. Different structures correspond to different levels of thermal abstraction, ranging from first-order single-zone models to multi-state configurations including internal thermal masses and envelope dynamics [10, 11]. However, despite extensive use in literature, systematic guidelines for selecting an appropriate $xR+yC$ structure remain limited. Model topology directly influences prediction accuracy, computational cost, parameter identifiability, and robustness under operational conditions [12, 13]. As structure complexity increases, parameter estimation becomes more challenging, and identifiability issues may arise [9, 14].

From a Digital Twin perspective, the selection of an RC structure is not merely a modelling choice, but a foundational design decision. The RC model effectively constitutes the first physical brick of an operational building DT. If the thermal core is poorly structured, subsequent control, optimisation, or predictive layers—such as Model Predictive Control (MPC)—inherit structural bias and uncertainty [15, 16]. Conversely, an adequately selected RC structure enables interpretable thermal state estimation, fast simulation, and scalable integration into O&M decision loops.

This paper investigates the selection and comparative evaluation of multiple RC thermal network structures for an existing institutional building [17]. The study focuses on Block A of a multi-block facility (Blocks A, B, and C), as illustrated in Fig. 1. One year of operational data—including indoor air temperature, outdoor temperature, and heating-related measurements—was used to calibrate and compare a hierarchy of $xR+yC$ configurations under consistent training, validation, and testing conditions. Rather than assuming a predefined structure, a systematic model bank was evaluated to analyse accuracy–complexity trade-offs, seasonal behaviour, and structural limitations. The dataset used in this study originates from the PLEIADa repository published in [17].

The contribution of this work is threefold. First, it positions RC modelling explicitly as the minimal thermal core of a building Digital Twin. Second, it formalises a structured comparison framework for $xR+yC$ configurations under

real operational data. Third, it critically analyses structural limitations, seasonal variability, and identifiability constraints, thereby clarifying the practical boundaries of reduced-order DT deployment in existing buildings.



Fig. 1. Pleiades building (University of Murcia), composed of three blocks (A, B, C). This study focuses exclusively on Block A for RC thermal modelling.

2 State of the Art

2.1 Digital Twin in the Built Environment: Definitions, Scope, and Coupling Levels

Digital Twin (DT) definitions vary across domains, so a coupling-based taxonomy was proposed by Kritzinger et al. to separate a Digital Model (DM), a Digital Shadow (DS), and a DT [1]. A DM was described as a model that does not use any form of automated data exchange [1]. A DS was described as an automated one-way link from the physical system to the digital representation, but not vice versa [1]. A DT was described as a bidirectionally coupled representation that is fully integrated in both directions [1].

In the built environment, DT reviews link DTs to continuous synchronization, analytics, and decision support beyond static Building Information Modeling (BIM) data [2, 3, 6, 18]. The move from BIM to DTs for Operation and Maintenance (O&M) was discussed by Lu et al., and DT capabilities were associated with updated data streams and model-based reasoning for operational decisions [6]. Energy efficiency is a central DT motivation in O&M, and O&M DT reviews classify applications such as monitoring, anomaly detection, predictive maintenance, operational optimization, and scenario analysis [4, 5]. Barriers were also reported across built-environment DT reviews, especially for interoperability, data integration, and incomplete digital documentation in existing buildings [2, 4, 5, 6].

2.2 RC-Network-Based Digital Twin: Fundamentals, $xR+yC$ Structure, and Model Construction

Resistance–Capacitance (RC) thermal networks are widely used as grey-box models, because a simplified physical structure is combined with parameter estimation from measured data [7, 8, 9]. Grey-box building models were reviewed by Li et al., and the grey-box position was described between white-box physics-based simulation and black-box statistical models [7]. Building thermal-network models were reviewed by Boodi et al., and RC configuration choices were shown to influence predictive performance and suitability for control [8].

Low-order identification was reviewed by Serasinghe et al., and the notation $xRyC$ (where x and y are integers) was summarized to describe RC model order using integer structure choices [9]. This notation supports the statement that a wall = $xR+yC$, because wall dynamics can be represented by selecting the number of lumped resistances and capacitances [9]. Low-order wall modelling was derived analytically by Ramallo-González et al., and a second-order formulation consisting of three resistors and two capacitors was presented, which corresponds to a 3R2C structure [10]. The REHVA overview of EN ISO 52016-1 described simplified dynamic methods and stated that the method includes the influence from technical building systems, control aspects and boundary conditions, which motivates disturbances and system coupling even in reduced-order models [19].

Model structure selection is a key issue because topology controls the trade-off between prediction accuracy and computational cost. Single-zone grey-box structures were compared by Chen et al., and oversimplified structures were reported to dominate error, while more complex structures increased runtime [12]. A selection procedure was proposed by Bacher and Madsen, where a hierarchy of candidate models was compared and statistical tests were used to identify a suitable model for a specific building and dataset [11]. Reviews also report inconsistent naming and structure descriptions across envelope-, zone-, and whole-building RC models, which complicates systematic comparison of $xR+yC$ choices [7, 8, 9].

Identification and calibration determine whether an RC structure becomes a reliable predictive model. Serasinghe et al. reported that initialization becomes challenging as structure complexity increases [9]. Bayesian calibration was applied by Marty-Jourjon et al. to several RC structures, and interpretability limits were discussed under strong lumping, while parameter distributions were estimated to represent uncertainty [14]. Deterministic and stochastic calibration were compared by Rouchier et al., and modelling mismatch was represented through process noise, which affected parameter robustness and forecasting stability [20]. Identifiability was analysed by Marty-Jourjon et al. for an ISO 13790-based RC model, and the limitation that only global resistance is identifiable was reported in their configuration [13]. Beyond linear models, non-linear grey-box formulations were introduced by Thilker et al. using Stochastic Differential Equations (SDEs) with additional states for heating system dynamics, and these models were positioned as predictive cores for advanced control [21].

2.3 DT-RC for Building Energy Optimisation: Model Predictive Control and Robustness

Model Predictive Control (MPC) computes control actions by optimizing predicted behaviour over a future horizon under constraints, and building MPC commonly targets energy or cost reduction under comfort constraints [15, 16]. RC grey-box models are often used for MPC because fast prediction is enabled while interpretable thermal states and parameters are retained [7, 8, 12]. MPC was implemented on a real building heating system by Široký et al., and energy savings of 15.54% and 16.94% were reported relative to reference strategies under their tested conditions [15]. Forecast uncertainty in residential control with Photovoltaic (PV) generation and a Battery Energy Storage System (BESS) was studied by Hu et al., and an improved double-layer MPC was proposed, with reported reductions of 15.12% in electricity cost and 10.50% in energy consumption compared with a baseline controller [16]. Bayesian and stochastic calibration studies provide complementary mechanisms by representing parameter uncertainty and modelling mismatch, which supports repeated updating and more robust forecasting in O&M settings [14, 20].

2.4 Critical Synthesis and Research Gaps

O&M DT reviews emphasize decision support for energy efficiency, but deployment is limited by integration barriers and incomplete documentation in existing buildings [2, 4, 5, 6]. RC grey-box networks provide a lightweight modelling basis because computational efficiency is combined with physical structure and calibration from measurements [7, 8, 9]. Gaps remain in (i) prescriptive $xR+yC$ structure selection under accuracy–runtime constraints, given the lack of standardized configurations [8, 11, 12]; (ii) identifiability under lumping and parameter correlation, which can restrict stable updating and physical meaning [14, 13]; and (iii) uncertainty-aware calibration and control robustness, because modelling mismatch and forecast errors influence forecasting stability and optimisation outcomes [20, 16]. These gaps motivate approaches that connect structure choice, identifiability, and uncertainty modelling to reliable MPC-driven optimisation in an operational DT-RC loop [9, 11, 12, 14, 20, 15, 16]. The present paper can be positioned to formalize $xR+yC$ structure selection by linking structure complexity to the documented accuracy–runtime trade-offs and the absence of standardized configurations.

3 Methodology and Data

3.1 Dataset Description and Building Scope

The experimental data used in this study originate from the PLEIADa dataset published by Martínez Ibarra et al. [17]. The dataset contains synchronized measurements of HVAC electrical consumption, indoor environmental variables, and meteorological data collected at the Pleiades building (University of Murcia, Spain) over approximately one year.

The Pleiades building is composed of three blocks (A, B, and C). This study focuses exclusively on Block A, as illustrated in Fig. 1. All signals associated with Block A were extracted using the metadata structure provided in [17].

It is important to emphasize that PLEIADData is not a raw Building Management System (BMS) dump. The dataset authors performed preprocessing including timestamp synchronization, signal alignment, metadata documentation, and consistency validation. This preprocessing significantly improves reproducibility and reduces common issues such as sensor misalignment or undocumented naming conventions.

3.2 Signal Construction and Preprocessing

Three core signals are constructed: the indoor air temperature $T_{in}(t)$, obtained through aggregation of indoor sensors; the outdoor temperature $T_{out}(t)$ from the on-site weather station; and the HVAC electrical consumption $P_{HVAC}(t)$ aggregated at block level.

Because direct thermal power injection is not measured, HVAC electrical consumption is mapped to an effective thermal input, as defined in Eq. (1):

$$Q(t) = \alpha P_{HVAC}(t) \quad (1)$$

where α is an unknown gain parameter representing effective thermal delivery efficiency.

Interpretation and limitations of the HVAC-to-thermal mapping. Because direct delivered thermal power is not available in the dataset, $P_{HVAC}(t)$ is mapped to an effective thermal input through a constant gain α . This gain should not be interpreted as a fixed physical efficiency: it implicitly aggregates multiple mechanisms such as system COP/efficiency variations with outdoor conditions, HVAC operating modes and control cycles, distribution losses, and auxiliary consumption. Consequently, α may partially compensate for envelope parameters (R/C) during calibration, which introduces practical parameter ambiguity and can contribute to regime-dependent bias. In particular, seasonal changes in HVAC effectiveness that are not represented by a constant α provide a plausible explanation for the observed late-year drift when a single global parameter set is used. This simplifying assumption is nevertheless consistent with the deployment objective of this paper: assessing parsimonious RC structures as a first DT thermal core under limited operational signals, while explicitly reporting the resulting limitations.

All signals are resampled to an hourly resolution ($\Delta t = 3600$ s), which captures dominant building thermal dynamics while attenuating high-frequency noise [8].

Indoor temperature aggregation. The PLEIADData repository provides multiple indoor temperature sensors distributed across Block A, measured at sub-hourly resolution and preprocessed by the dataset authors [17]. Let N_s denote the number of available indoor temperature sensors associated with Block A. At each time step, the indoor air temperature used for RC calibration is computed as the arithmetic mean across all available sensors, as defined in Eq. (2):

$$T_{in}(t) = \frac{1}{N_s(t)} \sum_{i=1}^{N_s(t)} T_{in}^{(i)}(t) \quad (2)$$

where $N_s(t)$ accounts for sensors available at time t after synchronization and cleaning.

No spatial weighting was applied, as detailed geometric information (surface areas, zone volumes, orientations) was not available. This aggregation therefore represents an effective block-level thermal state consistent with the lumped single-zone RC formulation adopted in this study. Missing values were implicitly handled through the synchronized dataset provided by the authors [17], and hourly resampling further attenuates high-frequency inter-zone variability.

It is important to note that this averaging procedure does not guarantee strict physical equivalence with a single well-mixed thermal zone. Instead, the resulting $T_{in}(t)$ should be interpreted as an effective thermal indicator of the dominant block-level dynamics, which may partly explain non-Gaussian residuals and seasonal bias observed in later sections.

3.3 Chronological Data Splitting

To emulate realistic Digital Twin deployment conditions and avoid temporal leakage, a chronological split is adopted: a training set from January 1 to September 30 (6460 samples), a validation set from October 1 to November 30 (1464 samples), and a test set from December 1 to December 31 (680 samples). The training set is used for parameter estimation, the validation set for structure selection and complexity penalization, and the test set for out-of-sample performance evaluation.

3.4 RC Thermal Network Formulation

Block A is modeled using lumped-parameter Resistance–Capacitance (RC) networks. For an $xR+yC$ structure, the continuous-time dynamics are expressed as Eq. (3):

$$C\dot{T}(t) + KT(t) = G_{out} T_{out}(t) + G_q Q(t) \quad (3)$$

with $C = \text{diag}(C_1, \dots, C_y)$, K derived from thermal resistances, and $T(t)$ the node temperatures. The corresponding state-space form is given in Eq. (4):

$$\dot{x}(t) = A(\theta)x(t) + b_{out}(\theta)T_{out}(t) + b_q(\theta)Q(t) \quad (4)$$

with the parameter vector defined in Eq. (5):

$$\theta = \{R_1, \dots, R_x, C_1, \dots, C_y, \alpha\} \quad (5)$$

Exact discretization is performed via the matrix exponential, as defined in Eq. (6):

$$A_d = e^{A\Delta t} \quad (6)$$

ensuring numerical stability for stiff thermal systems.

3.5 Parameter Estimation

Parameters are estimated by minimizing the quadratic prediction error over the training set, as defined in Eq. (7):

$$J(\theta) = \frac{1}{N} \sum_{k=1}^N \left(\hat{T}_{in,k} - T_{in,k} \right)^2 \quad (7)$$

subject to the positivity constraints in Eq. (8):

$$R_i > 0, \quad C_j > 0 \quad (8)$$

Optimization is conducted in log-space to enforce positivity and improve conditioning.

3.6 Implementation Details for Reproducibility

To ensure executable reproducibility, parameter estimation is implemented using the minimize routine from `scipy.optimize` with the bound-constrained quasi-Newton method L-BFGS-B. All parameters are optimized in log-space to enforce positivity: $\phi = \log(\theta)$, with $\theta = \{R_1, \dots, R_x, C_1, \dots, C_y, \alpha\}$.

Initialization is fixed and identical across candidate structures: $R_i = 0.2$, $C_j = 10^7$, and $\alpha = 10^{-4}$ (mapped to log-space before optimization). Wide but physically plausible bounds are enforced in log-space: $\log(R_i) \in [-10, 5]$, $\log(C_j) \in [8, 25]$, and $\log(\alpha) \in [-20, 2]$. The objective minimized over the training set is the mean squared error (MSE) on T_{in} . These implementation choices do not alter the model class but directly affect convergence behaviour and the robustness of structure comparison under real operational data.

Convergence was assessed based on the relative change in the objective function and parameter vector between successive iterations, as handled internally by the L-BFGS-B implementation in SciPy. Default stopping tolerances were retained to ensure consistent treatment across candidate structures.

A single deterministic initialization was used for each structure to ensure consistent comparison across the model bank. While multi-start strategies can provide additional insights into parameter dispersion and local minima sensitivity, the objective here is comparative structure ranking under identical optimization settings rather than exhaustive global parameter exploration. The relatively wide log-space bounds were intentionally selected to avoid artificially constraining structural expressiveness, while preventing non-physical parameter values and numerical instability.

3.7 RC Structure Families and Combinations

Two distinct structural families are evaluated.

LADDER structures. The LADDER family corresponds to canonical series RC chains with increasing order, defined in Eq. (9):

$$LADDER_n = nR \ nC \quad (9)$$

with $n = 1, \dots, 10$. These represent purely hierarchical lumped envelope models with increasing internal thermal mass representation.

STD structures (ISO-inspired). The STD family corresponds to physically structured topologies inspired by simplified ISO and grey-box modeling practice: 1R1C; 2R1C with parallel losses; 2R2C including air mass; 3R2C with air shunt; 3R3C two-mass series; 4R3C two masses plus air shunt; and 5R3C and higher multi-mass variants with shunt paths.

For clarity and reproducibility, each STD label (e.g., 4R3C with two masses and an air shunt) refers to a single, fixed connectivity between nodes (air node, envelope mass nodes, and direct loss/shunt path) as in Fig. 2. Thus, structure selection is performed over fully specified state-space models rather than over ambiguous model name categories.

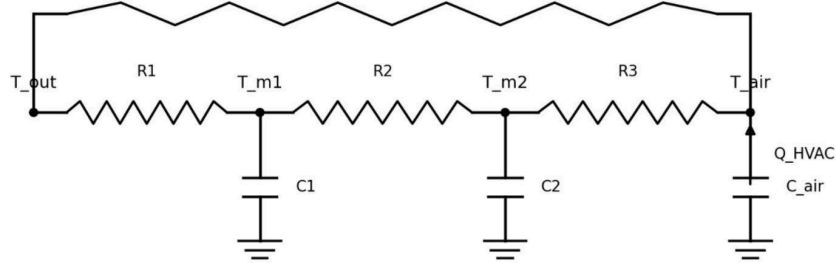


Fig. 2. Schematic of the grey-box RC thermal network (four resistors, three capacitors).

3.8 Model Comparison Criteria

All candidate structures are evaluated on the validation set using RMSE, MAE, the Akaike Information Criterion (AIC), and the Bayesian Information Criterion (BIC). The BIC is computed as defined in Eq. (10):

$$BIC = n \ln\left(\frac{RSS}{n}\right) + k \ln(n) \quad (10)$$

where k is the number of estimated parameters. Because performance metrics can vary substantially across seasons and operating regimes, all validation and test results are interpreted in a regime-aware manner, and no single period is treated as representative of year-round behaviour.

In this work, information criteria are used as practical parsimony scores for structure selection rather than as strict likelihood-based model evidence. Because validation data are used to mitigate overfitting (parameters identified on training, scores evaluated on validation) and because residuals may deviate from the i.i.d. Gaussian assumption under real operational conditions, AIC/BIC should be interpreted here as BIC-like penalized error criteria that complement RMSE/MAE in guiding an accuracy–complexity compromise.

4 Results and Discussion

4.1 Model Selection Across RC Structures

Nineteen candidate RC structures were evaluated, including ten canonical ladder configurations (1R1C to 10R10C) and nine ISO-inspired structured models incorporating air nodes and shunt paths.

Model comparison was performed using validation RMSE and information criteria (AIC/BIC). The best-performing model according to BIC was the STD 4R3C two masses plus air shunt structure, with a validation RMSE of 4.68 °C, a validation BIC of 4578.6, and a test RMSE of 0.86 °C. Structure selection is strictly based on validation performance and penalized complexity criteria; the December test results are reported solely for complementary regime-specific assessment.

In contrast, simple 1R1C structures exhibited higher validation RMSE (5.01 °C) and larger BIC values, despite having fewer parameters. Importantly, increasing structural complexity beyond 4R3C did not yield systematic improvements. Higher-order ladder models (up to 10R10C) showed nearly identical validation RMSE (5.01 °C) but significantly increased BIC values due to parameter penalization.

This suggests that, for the present building and the considered measurements (hourly T_{in} , T_{out} , and an effective HVAC input), additional RC states beyond a moderate order do not yield systematic validation gains, while increasing the risk of over-parameterization under limited excitation.

4.2 Full-Year Predictive Behaviour

Fig. 3 presents measured versus estimated indoor temperature over the full year for the selected 4R3C model. Two distinct regimes are observed: from January to mid-summer, the model tracks the dominant seasonal trend reasonably well; from late summer onward, a systematic positive bias progressively appears.

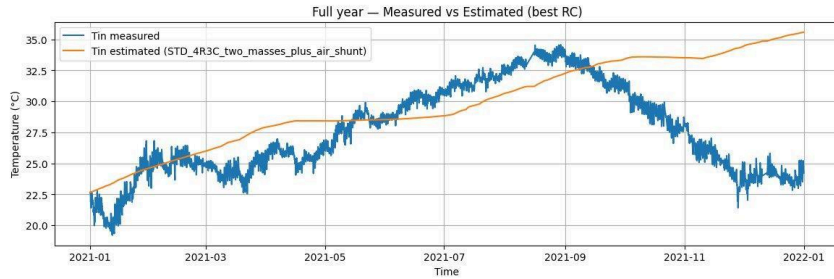


Fig. 3. Full-year measured versus estimated indoor temperature for the selected 4R3C RC model.

The gradual divergence suggests that a single global parameter set struggles to simultaneously represent winter and late-year dynamics. The deviation appears as a structural bias rather than high-frequency noise, indicating modelling mismatch rather than purely stochastic error.

4.3 Out-of-Sample Performance (December Test Set)

Fig. 4 shows the model performance restricted to the December test period. Despite the seasonal drift observed over the full year, short-horizon tracking

remains relatively stable, with a test RMSE of approximately $0.86\text{ }^{\circ}\text{C}$. The model captures short-term fluctuations but slightly underestimates daily variability amplitude. This behaviour suggests that dominant thermal inertia is captured, while finer control modulation or internal gain dynamics remain unmodelled.

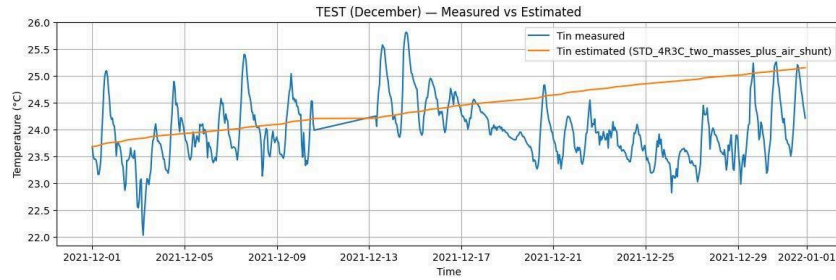


Fig. 4. Measured versus estimated indoor temperature during the December test period.

On the validation–test RMSE gap and its interpretation. The marked difference between the validation RMSE (Oct–Nov) and the December test RMSE should not be interpreted as evidence of systematically superior out-of-sample generalization. First, RMSE is scale-dependent and can decrease mechanically when indoor temperature variability is lower over a given period, even if the underlying modelling error mechanisms remain unchanged. Second, Oct–Nov frequently correspond to transitional operating conditions (e.g., shifts in setpoints, occupancy schedules, or HVAC regimes), which are structurally harder to capture using a single linear, lumped RC parameter set identified on earlier months. Third, the December period is interpreted here as a regime-specific robustness check under more stable winter operation, rather than as a comprehensive assessment of year-round predictive validity. Accordingly, full-year behaviour and residual structure (seasonal bias) remain the primary evidence used to discuss modelling limitations and practical DT deployment boundaries.

In addition, the indoor temperature variability during December is qualitatively lower than during the October–November transitional period. Since RMSE is directly influenced by the amplitude of the target signal, a reduction in variance mechanically reduces the absolute error magnitude, even when the structural modelling limitations remain unchanged. This scale effect further explains the apparent validation–test discrepancy without implying superior structural generalization.

4.4 Error Distribution Analysis

Fig. 5 presents the distribution of prediction errors over the full dataset. The error distribution is non-Gaussian and exhibits positive skewness during late-year periods. The presence of heavy positive tails confirms the systematic bias identified in Fig. 3. The absence of a symmetric zero-centered distribution indicates that residuals are not purely stochastic but reflect structural modelling limitations.

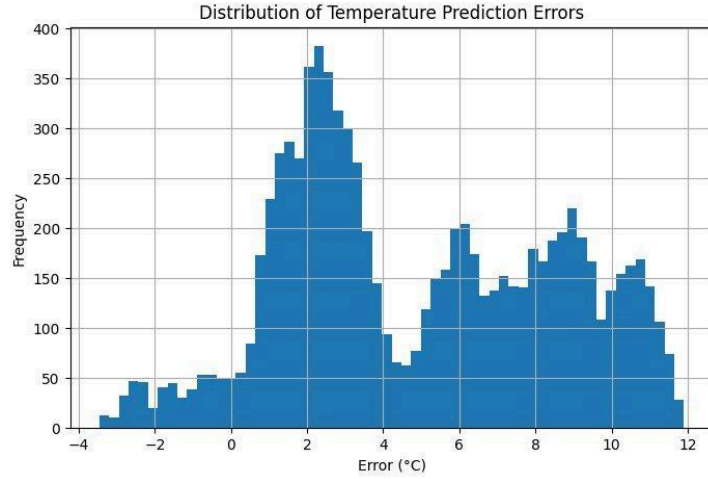


Fig. 5. Distribution of indoor temperature prediction errors over the full dataset.

4.5 Complexity Versus Performance Trade-Off

A key finding is that structural complexity alone does not guarantee improved predictive performance. Although higher-order ladder models introduce additional resistances and capacitances, validation RMSE remains nearly unchanged beyond low-order structures, while BIC increases significantly. The 4R3C structured model achieves the best compromise because it separates air and envelope thermal masses, it introduces a shunt pathway capturing direct heat exchange, and it avoids excessive deep mass states. This confirms that physically meaningful topology is more important than simply increasing model order.

4.6 Seasonality and Global Parameter Limitations

The observed seasonal drift highlights a fundamental limitation of global RC parameter identification over a single annual dataset. Possible explanations include season-dependent effective heat transfer coefficients, occupancy and internal gains variation, HVAC operational regime changes, and solar and envelope dynamics not explicitly modelled. A single global RC parameter set may therefore be insufficient to represent year-round dynamics in real buildings.

4.7 Implications for Digital Twin Construction

Despite imperfections, the selected RC structure provides interpretable physical parameters, computational efficiency, and acceptable short-term predictive accuracy. These characteristics confirm that RC networks can serve as a viable first modelling brick in a Digital Twin architecture when full BIM-based models are unavailable. The results emphasize that robustness, interpretability, and structural parsimony are critical for operational deployment.

4.8 Structural Identifiability and Absence of Geometric Information

A fundamental limitation of the present study lies in the absence of detailed geometric information, particularly the 2D architectural plan of Block A. In the absence of wall-by-wall surface areas, orientations, and construction layers, it is not possible to assign individual resistances and capacitances to specific envelope components (north wall, south façade, roof slab). Consequently, the identified RC parameters must be interpreted as effective lumped properties representing aggregated thermal behaviour rather than physically separable wall elements.

This limitation directly affects structural identifiability. Multiple parameter combinations may produce similar indoor temperature responses, especially under limited excitation conditions. As discussed in grey-box identification literature, lumping increases parameter correlation and reduces direct physical interpretability.

However, this constraint is also aligned with the objective of this study: to evaluate RC networks as a minimal first modelling brick for Digital Twin construction under limited data availability. In contexts where BIM models or detailed architectural plans are unavailable—common in existing building stock—lumped RC representations remain a viable and computationally efficient modelling strategy. Future work should therefore investigate hybrid approaches combining RC identification with geometric constraints derived from 2D plans or BIM models in order to reduce parameter ambiguity and improve seasonal robustness.

5 Conclusion

This study investigated RC grey-box thermal networks as a lightweight modelling layer for operational building Digital Twins under limited documentation. Using one year of PLEIAData measurements for Block A of the Pleiades building, a bank of nineteen candidate $xR+yC$ structures (ladder and ISO-inspired STD families) was calibrated on a chronological training set and compared on a separate validation period using accuracy metrics and penalized parsimony criteria.

For the studied building and dataset, results show that structural complexity alone does not guarantee improved predictive performance when available excitations and measured disturbances are limited. A structured 4R3C topology achieved the most favourable accuracy–parsimony compromise among the tested candidates, suggesting that physically meaningful connectivity can be more beneficial than increasing order through deep ladder states.

Full-year evaluation highlighted seasonal drift when a single global parameter set is used across multiple operational regimes. While short-horizon tracking during December yielded a lower RMSE, this should be interpreted as a regime-specific outcome under more stable winter operation rather than as a guarantee of improved generalization; the observed annual bias indicates that regime-dependent behaviour (e.g., HVAC operation changes, internal gains, and unmodelled disturbances) can dominate errors in lumped linear RC models.

The absence of detailed geometric information (e.g., 2D plans) further restricts physical interpretability: identified parameters should be interpreted as effective lumped properties rather than component-level quantities. Future work should

therefore consider regime-dependent calibration and the integration of additional disturbances (e.g., solar/internal gains proxies) and geometric constraints to reduce lumping ambiguity, enabling more robust DT integration and MPC-oriented applications.

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